

SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY (A)
RVS NAGAR, CHITTOOR

I B.Tech-I Semester:

Subject name: – Linear Algebra and Calculus
 Branch: Common to all branches

Sessional Examination-I
 Subject code: 23AHS04
 Max Marks: 25

Time: 2 Hours

Part A

Answer all the questions, each question carry equal marks

5 x 2 = 10 M

- 1 Define Echelon form of a matrix.
- 2 Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$
- 3 What is the consistency and Inconsistency of system of the linear equations?
- 4 State Cayley Hamilton theorem
- 5 Write the quadratic form of the matrix $= \begin{bmatrix} 2 & 0 & -2 \\ 0 & 2 & 1 \\ -2 & 1 & -2 \end{bmatrix}$.

Part B

Answer all the questions, each question carry equal marks

3 x 5 = 15 M

- 6.(a) Reduce the matrix into echelon form and hence find its

$$\text{rank} = \begin{bmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{bmatrix}$$

OR

- 6.(b) Find the inverse of $A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 4 & 5 \end{bmatrix}$ using gauss Jordan method.

- 7.(a) Solve the system of equations $x + y + z = 6$, $x - y + 2z = 5$, $3x + y + z = 8$,
 $2x - 2y + 3z = 7$.

OR

- 7.(b) Find the Eigen values and Eigen vectors of the matrix $\begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$

- 8.(a) Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$ Hence find A^{-1} and A^4 .

OR

- 8.(b) Reduce the Q.F. $10x_1^2 + 2x_2^2 + 5x_3^2 + 6x_2x_3 - 10x_3x_1 - 4x_1x_2$ to the canonical form through orthogonal transformation. Find a set of values of x_1, x_2, x_3 which will make the form vanish.

Handwritten notes for Q. 8(b):

$$S_1 = 7$$

$$S_2 = 11$$

$$S_3 = 0$$

Handwritten augmented matrix for Q. 8(b):

$$\left(\begin{array}{ccc|ccc} 1 & 4 & 5 & 0 & 0 & 1 \\ 0 & 1 & 3/2 & 0 & 1/6 & 1/3 \\ 0 & -10 & -14 & 1 & 0 & -3 \end{array} \right)$$

I B.Tech-I Semester:

Subject name: – Engineering Physics

Branch: EEE, ECE, IT, CAI, CSC, IOT, AI, AI&DS, CSBS

Sessional Examination-I

Subject code: 23AHS05

Time: 2 Hours

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 = 10 M

- 1 State the principle of superposition.
- 2 Recall any two differences between Fraunhofer and Fresnel diffraction.
- 3 Define diffraction grating.
- 4 Illustrate space lattice.
- 5 What is the range of X-ray wavelength.

Part B

Answer all the questions, each question carry equal marks

3 x 5 = 15 M

- 6.(a) Discuss the theory behind the interference of light due to thin films by reflection with suitable ray diagram.

OR

- 6.(b) Discuss the Fraunhofer diffraction due to single slit and build the necessary conditions for the intensity distribution on the screen.

- 7.(a) With suitable sketches, explain the various types of polarization's and their applications.

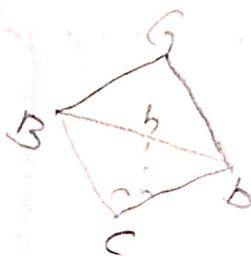
OR

- 7.(b) With suitable sketches, explain the Bravais lattices.

- 8.(a) Define Miller indices. Obtain the procedure to find the Miller indices of crystal plane. Discuss the characteristics of Miller Indices.

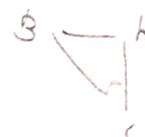
OR

- 8.(b) Describe the construction and working of Laue's X-ray diffraction for single crystal. List any two demerits of this method.



$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

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I B.Tech-I Semester: -ECE PART]

Subject name: – Basic Electrical and Electronics Engineering

Branch: EEE, IT, CAI, IOT, AI, AI&DS

Time: 2 Hours

Sessional Examination-I

Subject code: 23AEE01

Max Marks: 25

Part A

Answer all the questions, each question carry equal marks

5 x 2 = 10 M

- 1 What is the important application of a Zener Diode.
- 2 Draw the symbols of PNP and NPN transistors.
- 3 What is the advantage of Bridge rectifier over a center tapped full wave rectifier?
- 4 Mention the basic application of amplifiers.
- 5 What is a "unit" of electricity?

1 kWh

Part B

Answer all the questions, each question carry equal marks

3 x 5 = 15 M

- 6.(a) What is PN junction? Explain the formation of PN Junction.

OR

- 6.(b) Explain voltage current characteristics of a PN junction diode with suitable diagrams.

- 7.(a) What is Zener effect? Explain how the Zener diode can be used as a voltage regulator with necessary circuit diagram.

OR

- 7.(b) Explain the operation of Bridge rectifier with circuit diagram. Draw the input and output wave forms.

- 8.(a) Draw the symbol of OR and NOT, NOR, NAND, X-OR, and
X. NOR Gates and truth table.

- 8.(b) Discuss the causes and preventive measures of electric shock.

(c)

Explain the number system in Digital electronic.



SRI VENKATESWARA COLLEGE OF ENGINEERING AND TECHNOLOGY
R.V.S. Nagar, Chittoor-517127

DEPARTMENT OF MECHANICAL ENGINEERING

Engineering Graphics (23AME01)

Sessional Examination – I

Date: 15/11/2025

Max. Marks : 15

Year & Sem: I Year, I Sem

Time: 12.15 to 2.15 PM

Branch : IT, IOT, EEE

Answer all the questions, each question carries equal marks

1. (A) Construct a parabola whose distance of focus from directrix is 70 mm. The eccentricity is 1. Also draw a tangent and normal at 55 mm on the curve. (CO-1, APPLY)

(OR)

- (B) Draw a cycloid of a circle of 50 mm diameter, rolls along a straight line without slipping. (CO-1, APPLY)

2. (A) Draw an involute of circle of diameter 40 mm.. Also draw a tangent and normal to the curve at a distance 90 mm from the centre of the circle. (CO-1, APPLY)

(OR)

- (B) Draw the projections of a 60 mm long straight line, in the following positions.(CO-2, APPLY)

(a) Parallel to and 30 mm in front of the V.P and in the H.P.

(b) Perpendicular to the H.P, 30 mm in front of the V.P and its one end 20mm above the H.P.

(c) Inclined at 45° to the V.P, in the H.P. and it's one end is 30 in front of the V.P

3. (A) Draw the projections of the following points; keeping the distance between the projectors as 30 on the same reference line. (CO-2, APPLY)

a. A - 35 above H.P and on V.P.

b. B - 25 below H.P and 40 in front of V.P.

c. C - 20 below H.P and 30 behind V.P.

d. D - 20 above H.P and 30 behind V.P.

e. E - 35 above H.P and 30 in front of V.P.

(OR)

- (B) A line AB of 70 long is inclined at 45° to H.P and 30° to V.P. Its end A is on H.P and 30 in front of V.P. Draw its projections. (CO-2, APPLY)